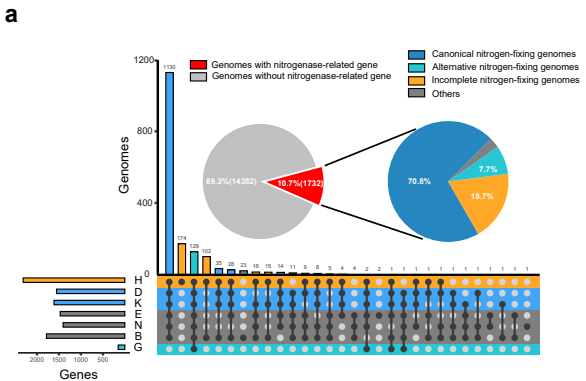
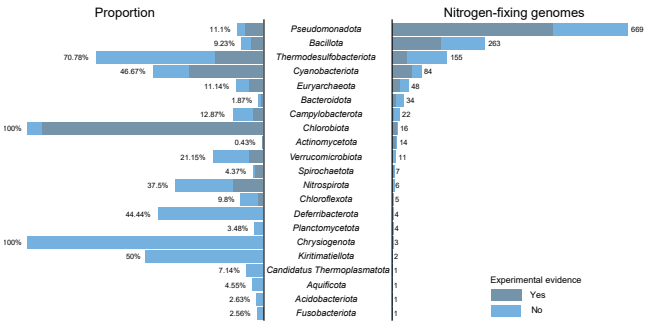

a
Genomes with nitrogenase-related gene
Genomes without nitrogenase-related gene
Canonical nitrogen-fixing genomes
Alternative nitrogen-fixing genomes
Incomplete nitrogen-fixing genomes
Others
Genomes
1200
800
400
0
1130
174
129
102
35
28
23
16
15
14
11
9
8
5
4
4
2
2
1
1
1
1
1
1
1
1
1
1
1
89.3%(14382)
10.7%(1732)
70.8%
7.7%
18.7%
H
D
K
E
N
B
G
2000
1500
1000
500
Genes


c
Proportion
Nitrogen-fixing genomes
11.1%
Pseudomonadota
669
9.23%
Bacillota
263
70.78%
Thermodesulfobacteriota
155
46.67%
Cyanobacteriota
84
11.14%
Euryarchaeota
48
1.87%
Bacteroidota
34
12.87%
Campylobacterota
22
100%
Chlorobiota
16
0.43%
Actinomycetota
14
21.15%
Verrucomicrobiota
11
4.37%
Spirochaetota
7
37.5%
Nitrospirota
6
9.8%
Chloroflexota
5
44.44%
Deferribacterota
4
3.48%
Planctomycetota
4
100%
Chrysiogenota
3
50%
Kiritimatiellota
2
7.14%
Candidatus Thermoplasmatota
1
4.55%
Aquificota
1
2.63%
Acidobacteriota
1
2.56%
Fusobacteriota
1
Experimental evidence
Yes
No

b
Tree scale
0.2
Phylum
Actinomycetota
Bacillota
Bacteroidota
Campylobacterota
Cyanobacteriota
Other phylum
Euryarchaeota
Mycoplasmatota
Pseudomonadota
Spirochaetota
Thermodesulfobacteriota
Genome classification
Canonical nitrogen-fixing genomes
Alternative nitrogen-fixing genomes
Incomplete nitrogen-fixing genomes
Others